

Kurt Braunlich

Address National Institute of Mental Health
Laboratory of Brain and Cognition
4C216 Building 10
Bethesda, MD 20892

Email kurt.braunlich@nih.gov
Website kbraunlich.github.io
Phone (360) 535-3495

Education

- 2015 **Colorado State University**
Ph.D., Cognitive Psychology.
Advisor: Dr. Carol Seger
Dissertation: Categorical Evidence, Confidence, and Urgency during the Integration of Multi-Feature Information
- 2010 **Western Washington University**
M.S., Experimental Psychology.
Advisor: Dr. Kristi Lemm
Thesis: Thought in the Absence of Attention
- 2004 **Catholic University of America**
B.A., Cum laude with honors. Experimental Psychology.
Advisor: Dr. Carol Glass

Research

- 2019– **National Institute of Mental Health**
Research Fellow, 2024–; Postdoctoral Fellow, 2019–24.
Supervisors: Dr. Chris Baker & Dr. Bevil Conway
Project: How accurate is the macaque model of the human brain? Exploring shared and species-specific multivariate representational structure using fMRI
- 2016 - 19 **University College London**
Postdoctoral Research Associate.
Advisor: Dr. Bradley Love
Project: Computational Modeling of Category Learning
- 2015 - 16 **Georgia Institute of Technology**
Postdoctoral Scholar.
Advisor: Dr. Mark Wheeler
Project: Neurobiological Mechanisms of Decision-Making
- 2010 - 15 **Colorado State University**
Graduate Student Researcher.
Advisor: Dr. Carol Seger
Project: Neurobiological Mechanisms of Categorization
- 2009 - 10 **Western Washington University**
Graduate Student Researcher.
Advisor: Dr. Kristi Lemm
Project: Thought in the Absence of Attention
- 2004 **National Institute of Mental Health: Clinical Brain Disorders Branch**
Research AssistantSpecial Volunteer.
Advisor: Dr. Terry Goldberg

Publications

- 2026 **Braunlich, K.**, Duyck, M., Duffield, S., Behel, K., Baker, C. I., & Conway, B. R. (2026). Translating functional brain activity between humans and monkeys. *bioRxiv*.
- 2025 Seger, C. A., **Braunlich, K.**, & Liu, Z. (2025). Integration of contextual reward motivation with category representations. *NeuroImage*, 323, 121589.
- Loggia, S., Duffield, S., **Braunlich, K.**, & Conway, B. (2025). Color and spatial frequency provide functional signatures of retinotopic visual areas. *The Journal of Neuroscience*, 45(2), e1673232024.
- 2022 **Braunlich, K.**, & Love, B. C. (2022). Bidirectional influences of information-sampling and concept learning. *Psychological Review*.
- 2019 Zeithamova, D., Mack, M. L., **Braunlich, K.**, Davis, T., Seger, C. A., van Kesteren, M., & Wutz, A. (2019). Brain mechanisms of concept learning. *The Journal of Neuroscience*.
- Braunlich, K.** & Love, B. C. (2019). Occipitotemporal representations reflect individual differences in conceptual knowledge. *Journal of Experimental Psychology: General*.
- Braunlich, K.**, Seger, C. A., Jentink, K. G., Buard, I., Kluger, B. M., & Thaut, M. H. (2019). Rhythmic auditory cues shape neural network recruitment in Parkinson's disease during repetitive motor behavior. *European Journal of Neuroscience*.
- 2017 **Braunlich, K.**, Liu, Z. & Seger, C. A. (2017). Occipitotemporal category representations are sensitive to abstract category boundaries defined by generalization demands. *The Journal of Neuroscience*, 37(32), 7631–7642.
- 2016 **Braunlich, K.** & Seger, C. A. (2016). Categorical evidence, confidence, and urgency during probabilistic categorization. *NeuroImage*, 125, 941-952.
- 2015 Seger, C. A., **Braunlich, K.**, Wehe, H., & Liu, Z. (2015). Generalization in category learning: The roles of representational and decisional uncertainty. *The Journal of Neuroscience*, 35(23), 8802–8812.
- Braunlich, K.**, Gomez-Lavin, J., & Seger, C. A. (2015). Frontoparietal networks involved in categorization and item working memory. *NeuroImage*, 107, 146–162.
- Liu, Z., **Braunlich, K.**, Wehe, H. S., & Seger, C. A. (2015). Neural networks supporting switching, hypothesis testing, and rule application. *Neuropsychologia*, 77, 19–34.
- Seger, C. A., & **Braunlich, K.** (2015). Category learning. In A. W. Toga (Ed.), *Human Brain Mapping: An Encyclopedic Reference* (Vol. 3, 487–492). Elsevier Inc.
- 2013 **Braunlich, K.**, & Seger, C. A. (2013). The basal ganglia. *Wiley Interdisciplinary Reviews: Cognitive Science*, 4(2), 135–148.

Invited Talks and Lectures

- 2026 Identifying functional homologies in macaque and human cerebral cortex. *Talk presented to the Laboratory of Neuropsychology, National Institute of Mental Health*.
- 2024 Project MACMAN: Establishing quantitative similarities between macaque and human brains. *Talk presented at the Julius Axelrod symposium at the National Institute of Health*.
- 2021 Exploring representational similarities in human and macaque using fMRI. *Talk presented to Laboratory of Sensation and Perception at the National Eye Institute*.
- 2020 Exploring putative homologies between Human and Monkey brains using fMRI. *Talk presented to Laboratory of Brain and Cognition at the National Institute of Mental Health*.

- 2016 Differentiating signals related to categorical evidence, confidence and urgency during probabilistic categorization. *Talk presented at the Center for Advanced Neuroimaging, the Georgia Institute of Technology.*

Presentations

Conference talks

- 2024 **Braunlich, K.** Comparing macaque and human functional organization using fMRI. *Talk presented at the Annual Meeting of the Society for Neuroscience.*
- 2010 **Braunlich, K.** Thought in the absence of attention. *Talk presented at the Northwest Cognition and Memory annual meeting.*

Posters

- 2026 **Braunlich, K.**, Duyck, M., Duffield, S., Behel, K., Baker, C. I., & Conway, B. R. Translating functional brain activity between humans and monkeys. *Poster presented at the Cognitive Computational Neuroscience annual meeting.*
- 2023 **Braunlich, K.**, Duyck, M., Behel, K., Duffield, S., Baker, C., & Conway, B. R. Project MACMAN: Establishing quantitative similarities between macaque and human brains. *Poster presented at the NIH 75'th Anniversary Symposium.*
- 2023 **Braunlich, K.**, Duyck, M., Behel, K., Duffield, S., Conway, B. R., & Baker, C. I. Exploring similarities and differences in human and macaque representational structure using fMRI. *Poster presented at the Visual Sciences Society annual meeting.*
- 2019 **Braunlich, K.**, & Seger, C. A. (2019). Integration of reward and representational information during categorical decision making in the angular gyrus and ventromedial prefrontal cortex. *Poster presented at the Society for NeuroEconomics annual meeting.*
- 2018 **Braunlich, K.**, & Seger, C. A. (2018). Goal-directed temporal modulation of probabilistic decision-making. *Poster presented at the Society for NeuroEconomics annual meeting.*
- 2017 **Braunlich, K.**, & Love, B. C. (2017). Occipitotemporal representations are modulated by conceptual knowledge and interact with a frontoparietal Network. *Poster presented at the Society for Neuroscience annual meeting.*
- Braunlich, K.**, & Love, B. C. (2017). Occipitotemporal representations are modulated by conceptual knowledge and interact with a frontoparietal network. *Poster presented at the Cognitive Computational Neuroscience annual meeting.*
- Braunlich, K.**, Jentink, K., Kluger, B. Thaut, M. H., & Seger, C. A. (2017). Rhythmic auditory cues shape neural network plasticity underlying motor control in Parkinson's disease. *Poster presented at the International Basal Ganglia Society annual meeting.*
- 2015 Thaut, M. H., **Braunlich, K.**, Kluger, B., McIntosh, G., & Seger, C. A. (2015). Beneficial effects of rhythmic auditory stimuli in the production of voluntary motor movements in Parkinson's disease. *Poster presented at the Society for Neuroscience annual meeting.*
- Braunlich, K.** & Seger, C. A. (2015). The accumulation of probabilistic categorical information. *Poster presented at the Organization for Human Brain Mapping annual meeting.*

- Braunlich, K.** & Seger, C. A. (2015). The accumulation of probabilistic categorical information. *Poster presented at the Cognitive Neuroscience Society annual meeting.*
- Seger, C. A. & **Braunlich, K.** (2015). Decision making and the strategic control of generalization. *Poster presented at the Cognitive Neuroscience Society annual meeting.*
- Braunlich, K.** & Seger, C. A. (2015). The accumulation of probabilistic categorical information. *Poster presented at the Front-Range Neuroscience Group annual meeting.*
- 2013 Dimond, A. J., **Braunlich, K.**, & Seger, C. A. (2013). fMRI and behavioral investigation of categorical evidence integration. *Poster presented at the Association for Psychological Sciences annual meeting.*
- 2012 **Braunlich, K.**, Fan, G., & Seger, C. A. (2012). The accumulation of information for categorization. *Poster presented at the Society for Neuroscience annual meeting.*
- Braunlich, K.**, Gomez-Lavin, J., & Seger, C. A. (2012). Shared and distinct frontal, parietal and striatal bases for item and category working memory. *Poster presented at the Organization for Human Brain Mapping annual meeting.*
- Gomez-Lavin, J., **Braunlich, K.**, Seger, C. A. & Horwitz, Z. (2012). Frontal-parietal network differences for item and category working memory. *Poster presented at the CURC Annual Conference at CSU.*
- Gomez-Lavin, J., **Braunlich, K.** & Seger, C. A. (2012). Frontal-parietal network differences for item and category working memory. *Poster presented at the Cognitive Neuroscience Society annual meeting.*
- Gomez-Lavin, J., **Braunlich, K.** & Seger, C. A. (2012). Frontal-parietal network differences for item and category working memory. *Poster presented at the Southern Society of Philosophy and Psychology annual meeting.*
- 2011 Spiering, B. J., **Braunlich, K.** & Seger, C. A. (2011). An fMRI study of the basal ganglia's role in multiple category-learning systems. *Poster presented at the Cognitive Neuroscience Society annual meeting.*
- Spiering, B. J., **Braunlich, K.**, & Seger, C. A. (2011). Switching between category learning systems: An fMRI Study. *Poster presented at the Executive Function and Emotion Conference at the University of Colorado, Boulder.*
- Spiering, B. J. **Braunlich, K.**, & Seger, C. A. (2011). Switching between category-learning systems: An fMRI Study. *Poster presented at the Molecular, Cellular and Integrative Neuroscience Conference at Colorado State University.*

Awards

- 2015 **David P. McCabe Award for Excellence in Research.** Colorado State University. Fort Collins, CO.
- Travel Award.** Colorado State University. Fort Collins, CO.
To attend the annual meeting of the Cognitive Neuroscience Society
- 2009 **Graduate Student Research Grant.** Western Washington University. Bellingham, WA.
\$600 to support my research concerning the development of preference in the absence of attention
- 2004 **Development Coach of the Year.** United States Olympic Committee. Colorado Springs, CO.
For my work with the US Whitewater Slalom Team

Teaching Experience

2017; 2018

Category learning. Guest lecture, University College London.

Lecture on category learning for master's students in Knowledge, Learning and Inference.

Fall 2013; Spring 2014

Cognitive Neuroscience Laboratory. Colorado State University, Fort Collins, CO.

Instructor for an undergraduate laboratory course (approx. 16 students) in Cognitive Neuroscience. The class focused on neuroanatomy, as well as the basic logic, theory, and methodology underlying the study of cognitive neuroscience. Class format included traditional lectures and lab-projects. The students gained hands-on experience with electrophysiological data collection and analysis (EEG & ERPs), fMRI data analysis, and dissections of post-mortem brains. Grades were based on quizzes, a final paper, and a final presentation.

Summer 2013; Fall 2014

Cognitive Psychology Laboratory. Colorado State University, Fort Collins, CO.

Instructor for an undergraduate laboratory course (approx. 16 students) in Cognitive Psychology. The class was designed to give students experience conducting and evaluating empirical research in cognitive psychology. Students collected data from each other using computer-based cognitive experiments, shared data with one another using google sheets, created simple plots using MATLAB, and learned how to conduct simple statistical tests. Grades were based on homework assignments, lab-reports, a final paper, and a final presentation.

Spring 2015

Biological Psychology Laboratory. Colorado State University, Fort Collins, CO.

Instructor for an undergraduate laboratory course (approx. 16 students) in Biological Psychology. The class explored interactions between biology, cognition, and behavior. Students learned about invasive and non-invasive neuroscience methods, and gained hands-on experience with measurement and analysis of electrodermal activity, heart-rate and electroencephalography. Grades were based on daily lab-reports, homework, and an independent project that culminated in a final presentation.

Spring 2015

Introduction to fMRI Analysis. Colorado State University, Fort Collins, CO.

Instructor for an undergraduate seminar (5 students) designed to provide an overview of fMRI data analysis for students considering a career in cognitive neuroscience. Students learned to use MATLAB and Statistical Parametric Mapping software (SPM) to perform preprocessing and GLM-based analysis.

Fall 2009

Research Methods Laboratory. Western Washington University, Bellingham, WA.

Instructor for an undergraduate laboratory course (approx. 15 students) designed to provide an introduction to basic statistical concepts and analyses. Students learned how to use SPSS to calculate descriptive statistics, *t*-tests, and one-way ANOVA's.

Teaching Assistantships

Fall 2010

Introduction to Psychology. Colorado State University, Fort Collins, CO.

Teaching assistant for an undergraduate course (approx. 30 students) designed to provide a general introduction to the field of psychology.

Spring 2010

Research Methods. Colorado State University, Fort Collins, CO.

Teaching assistant for an undergraduate course (approx. 20 students) designed to introduce students to basic descriptive and inferential statistics.

Spring, 2009

Introduction to Psychology. Western Washington University.

Teaching assistant for an undergraduate psychology course (approx. 20 students) designed to provide a general introduction to the field of psychology.

Fall, 2008

Experimental Approaches in Research. Western Washington University.

Teaching assistant for an undergraduate course (approx. 15 students) designed to introduce students to basic descriptive and inferential statistics.

Peer Review

• Brain Structure and Function • Communications Biology • Human Brain Mapping • Journal of Cognitive Neuroscience • Journal of Neuroscience • Memory and Cognition • NeuroImage • Psychological Review • Psychonomic Bulletin & Review • Scientific Reports

Programming

• Python • MATLAB • R • Julia